

Low-Light Image Enhancement with Cross-Dataset Generalization

Cosimo Curri
Politecnico di Bari
Bari, Italy

c.curri@studenti.poliba.it

Abstract

...

1. Introduction

...

2. Experimental Protocol

2.1. Datasets

The pipeline developed in this work is based on four datasets, each with a specific role in the experimental protocol: (i) LOL-v2 [4], (ii) LOL-v1 [3], (iii) MILs [2], and (iv) ExDark [1]. All models are trained on a single source domain and are then evaluated on additional datasets without fine-tuning. This choice is intended to measure cross-dataset generalization, i.e., the ability of each model to enhance low-light images whose distribution differs from the training data.

The main training dataset is LOL-v2. This dataset contains both a real-captured subset and a synthetic subset. In the main protocol, only the real-captured subset is used, meaning that the low-light/normal-light pairs are acquired from real scenes rather than obtained through synthetic low-light degradation. LOL-v2 is a paired dataset: each low-light image is associated with a corresponding normally exposed reference image, which can be used as the target for supervised training and evaluation. The official training split of the real-captured subset is used to train all models, while the official real-captured test split is kept for in-domain evaluation. In this context, “in-domain” means that training and testing images come from the same dataset and acquisition setting. The synthetic subset is kept separate from the main protocol and is used for an additional ablation study, aimed at evaluating whether synthetically degraded low-light examples improve or degrade generalization.

LOL-v1 is used only as a paired cross-dataset evaluation set. It represents a relatively close cross-dataset scenario,

because it belongs to the same general benchmark family as LOL-v2.

MILs is used as a second paired cross-dataset evaluation set. In particular, the low-resolution DSLR split, MILs-DSLR, is adopted to keep the evaluation computationally manageable. Unlike LOL-v1, MIL introduces a controlled illumination setting. Its input images are associated with different illumination levels, and each input has a corresponding well-illuminated reference image. This makes it possible to evaluate the models not only with global results over the whole dataset, but also by studying how performance changes as the input illumination becomes more severe.

Finally, ExDark is used as an unpaired stress-test dataset. Here, unpaired means that the dataset provides low-light images without a corresponding normally exposed reference image. Therefore, ExDark is not used for reference-based comparison, but for no-reference evaluation and qualitative failure analysis on real-world dark scenes. Its category-based organization is preserved, meaning that images remain grouped according to the object category provided by the dataset, such as cars, people, animals, or indoor objects. This organization may help inspect whether some failure cases are related to the type of scene or object.

3. Conclusion

...

References

- [1] Yuen Peng Loh and Chee Seng Chan. Getting to know low-light images with the exclusively dark dataset. *Comput. Vis. Image Underst.*, 178:30–42, 2019. 1
- [2] Maria Pilligua, David Serrano-Lozano, Pai Peng, Ramon Baldrich, Michael S. Brown, and Javier Vazquez-Corral. Evaluating low-light image enhancement across multiple intensity levels. *CoRR*, abs/2511.15496, 2025. 1
- [3] Chen Wei, Wenjing Wang, Wenhan Yang, and Jiaying Liu. Deep retinex decomposition for low-light enhancement. In *BMVC*, page 155. BMVA Press, 2018. 1
- [4] Wenhan Yang, Wenjing Wang, Haofeng Huang, Shiqi Wang, and Jiaying Liu. Sparse gradient regularized deep retinex

network for robust low-light image enhancement. *IEEE Trans. Image Process.*, 30:2072–2086, 2021. [1](#)